

Challenge (Habanero)

- Q1.** A hiker is packing energy bars for a trip. If she needs 12 bars for the first day and 15 bars for the second day, what is the least common multiple (LCM) of 12 and 15?
- (A) 30
(B) 36
(C) 60
(D) 72
(E) 90
- Q2.** A fuel tank can hold 24 gallons of gasoline. If a car uses 4 gallons to travel 10 miles, how many gallons will it use to travel 30 miles? What is the greatest common factor (GCF) of 24 and 4?
- (A) 1
(B) 2
(C) 4
(D) 6
(E) 8
- Q3.** In a forest, the population of rabbits is 48 and the population of deer is 18. What is the GCF of 48 and 18?
- (A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q4.** A park ranger needs to distribute food to monkeys. If she has 20 bags of food and each bag contains 6 servings, how many servings are there in total? What is the LCM of 20 and 6?
- (A) 20
(B) 30
(C) 60
(D) 120
(E) 180
- Q5.** A group of friends want to go on a hike that is 45 miles long. If they can hike 5 miles per day, how many days will it take them to complete the hike? What is the GCF of 45 and 5?
- (A) 1
(B) 3
(C) 5
(D) 9
(E) 15
- Q6.** A car travels 24 miles per gallon of gasoline. If the gas tank can hold 16 gallons, how many miles can the car travel on a full tank? What is the LCM of 24 and 16?
- (A) 48
(B) 64
(C) 96
(D) 128
(E) 192
- Q7.** A wildlife sanctuary has 50 animals in one enclosure and 25 animals in another. What is the GCF of 50 and 25?
- (A) 1
(B) 5
(C) 10
(D) 25
(E) 50
- Q8.** A hiker needs to climb 36 flights of stairs to reach the top of a mountain. If she can climb 4 flights per hour, how many hours will it take her to reach the top? What is the LCM of 36 and 4?
- (A) 12
(B) 18
(C) 36
(D) 48
(E) 72

Open Ended Questions (FRQ)

- Q9.** A group of friends want to go on a camping trip. They need to buy food for 18 people for 5 days. If each person eats 3 meals per day, how many meals will they need in total? Use GCF and LCM to find the total number of meals.
- Q10.** A car travels 120 miles per 4 gallons of gasoline. If the gas tank can hold 20 gallons, how many miles can the car travel on a full tank? Use GCF and LCM to find the total number of miles.

Chef's Tips

- Tip 1: Check for prime factorization
- Tip 2: Verify the units of measurement
- Tip 3: Encourage students to use diagrams and charts to visualize the problems